In [1]:
```
import pandas as pd
```

In [2]:
```
# TASK 1: INNER JOIN
users = pd.DataFrame({"user_id": ["U1", "U2", "U3", "U4", "U5"],"username": ["Ka
"city": ["Mumbai", "Delhi", "Pune", "Surat", "Vadodara"]})
```

In [3]:
```
orders = pd.DataFrame({"order_id": ["O1", "O2", "O3", "O4", "O5", "O6", "O7"],"u
"order_amount": [500, 750, 300, 1200, 450, 800, 650]})
```

In [4]:
```
result = pd.merge(users, orders, on="user_id", how="inner")

print("TASK 1")
print(result)
```

```
TASK 1
  user_id username    city order_id  order_amount
0      U1    Kahan  Mumbai       O1           500
1      U1    Kahan  Mumbai       O3           300
2      U1    Kahan  Mumbai       O7           650
3      U2     Kunj   Delhi       O2           750
4      U2     Kunj   Delhi       O5           450
5      U3     Dhir    Pune       O4          1200
6      U4   Vikas    Surat       O6           800
```

In [5]:
```
# TASK 2: LEFT JOIN
result = pd.merge(users, orders, on="user_id", how="left")

print("\nTASK 2")
print(result)
```

```
TASK 2
  user_id username      city order_id  order_amount
0      U1    Kahan    Mumbai       O1         500.0
1      U1    Kahan    Mumbai       O3         300.0
2      U1    Kahan    Mumbai       O7         650.0
3      U2     Kunj     Delhi       O2         750.0
4      U2     Kunj     Delhi       O5         450.0
5      U3     Dhir      Pune       O4        1200.0
6      U4    Vikas     Surat       O6         800.0
7      U5  Aditya  Vadodara      NaN           NaN
```

In [8]:
```
# TASK 3: OUTER JOIN
restaurants = pd.DataFrame({"restaurant_id": [101, 102, 103, 104],"name": ["Domi
"city": ["Mumbai", "Delhi", "Pune", "Surat"]})

reviews = pd.DataFrame({"review_id": [1, 2, 3, 4],"restaurant_id": [101, 102, 10
"reviewer_city": ["Mumbai", "Delhi", "Pune", "Pune"]})

result = pd.merge(restaurants,reviews,on="restaurant_id",how="outer")

print("\nTASK 3")
print(result)
```

```
TASK 3
   restaurant_id       name    city  review_id  rating reviewer_city
0            101    Dominos  Mumbai        1.0     5.0        Mumbai
1            102  McDonalds   Delhi        2.0     4.0         Delhi
2            103  Pizza Hut    Pune        3.0     5.0          Pune
3            104     Subway   Surat        NaN     NaN           NaN
4            105        NaN     NaN        4.0     3.0          Pune
```

In [11]:
```
# TASK 4: MERGE USING user_id AND city

playlists = pd.DataFrame({"playlist_id": ["P1", "P2", "P3", "P4"],"user_id": ["U
"genre": ["Pop", "Rock", "Bollywood", "Hip-Hop"],"city": ["Mumbai", "Delhi", "Su

user_profiles = pd.DataFrame({"user_id": ["U1", "U2", "U3", "U4"],"username": ["
"city": ["Mumbai", "Delhi", "Pune", "Vadodara"]})

result = pd.merge(playlists,user_profiles,on=["user_id", "city"],how="inner")

print("\nTASK 4")
print(result)
```

```
TASK 4
  playlist_id user_id    genre      city username
0          P1      U1      Pop    Mumbai    Kahan
1          P2      U2     Rock     Delhi     Kunj
2          P4      U4  Hip-Hop  Vadodara    Vikas
```

In [12]:
```
# TASK 5: CHATGPT / COPILOT - RIGHT JOIN
products = pd.DataFrame({"product_id": ["P1", "P2", "P3", "P4"],"name": ["Laptop
"category": ["Electronics", "Electronics", "Audio", "Wearable"]})

ratings = pd.DataFrame({"rating_id": ["R1", "R2", "R3", "R4", "R5"],"product_id"
"rating": [5, 4, 5, 3, 4]})
```

In [13]:
```
# Generated with ChatGPT and modified to RIGHT JOIN
result = pd.merge(
    products,
    ratings,
    on="product_id",
    how="right"
)

print("\nTASK 5")
print(result)
```

```
TASK 5
  product_id        name     category rating_id  rating
0         P1      Laptop  Electronics        R1       5
1         P2      Mobile  Electronics        R2       4
2         P1      Laptop  Electronics        R3       5
3         P5         NaN          NaN        R4       3
4         P3  Headphones        Audio        R5       4
```

In [ ]:
```
# I used ChatGPT to generate the pandas merge code.
# I changed how="inner" to how="right" to keep all records
# from the ratings DataFrame.
```